DNA, genes and chromosomes

Today I can… I can label DNA, a gene and a chromosome in a simple model. I can explain that a gene is a section of DNA with a protein instruction. I can distinguish one chromosome from the genome.

Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

Teacher and cover plan · 1–11

1. Outcome

I am learning to show how a gene is part of DNA and how DNA is packed into chromosomes. Weekly binding: Describe DNA, genes & chromosomes.

2. Arrival · 5 min

Private predictable start: notice one feature in string-and-clip DNA model. Point/pass allowed. No prior knowledge required.

3. Starter · 3 min

Hear each response. Keys: evidence checks treatment benefit/risk [A2W3]; each daughter-cell nucleus after mitosis [A2W1]; nucleus [A1W1]. Confirm actual teaching.

4. I do · 5 min

Words 2 + full model 3. Gene is a short DNA section coding for a protein; package DNA as chromosome within full DNA-set genome. Clip is marker. Show-me answer: gene is DNA section. Model not to scale.

5. We do · 7 min

Missing gene label 2 + match 3 + route rehearsal 2. Keys gene/DNA/chromosome/genome. Reset inside 7. F/M/H tasks and frames in pack; no filled independent answer.

6. You do · 8 min

Choose accurate Model A, explain one reason, then ONE route. Explain/question then swap, own responses. Optional prepared model and two-diagram audit within 8. Reset inside 8.

7. Routes and hinge · 5 min

We do 2 hinge B/second. Hear BOTH independent reasons. Both secure → transfer; re-model unsure learner + parallel genome hinge while other continues. Supported labels; Independent explains DNA section; Aspire corrects label/protein instruction; Go deeper distinguishes whole genome from gene.

8. Access

Any-language plan; point/sign/speech/AAC/draw/type/write/exact scribe/staff photo. No forced disclosure or handling. Pause/private correction; re-enter one label. Follow individual plan.

9. TA role

Use held five-step ladder in TA/answers, no physical prompts. Record deepest prompt, valid mode and independence separately; teacher owns hinge decision.

10. Exit · 3 min

Two prompts group source model/reason and feedback/next precise sentence; all criteria/goal/How I know/Audience retained. Each own response. No extra science task.

11. Evaluation · Review 4 min

Review includes reset, actual VF/NS(+), real edit // or read-back ?. Date taught _____ Criteria/adaptations/reasons/coverage/evidence _____. R judged next lesson only.

Made from the web lesson SCI-LAUNCH-A2-W04 · version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

DNA, genes and chromosomes

Today I can… I can label DNA, a gene and a chromosome in a simple model. I can explain that a gene is a section of DNA with a protein instruction. I can distinguish one chromosome from the genome.

Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

Teacher and cover plan · 12

12. Kit, safety, evidence and timing

Optional rehearsal: short DNA cord, large blunt clips, genome/chromosome frames and tray. Mark one gene with a clip and nest the DNA cord in frames. Clip is a marker only. Picture and cord lengths/shapes are not to scale; genome is all DNA, including other DNA. No medical claim about a learner.

Risks: clip pinch/loose cord catch. Use large blunt clips and short cord on a tray, away from necks. Short length only, counted out/back in, adult-supervised; remove at once if dysregulated. Confirm actual prepared kit/local school risk assessment. Digital labelled-picture alternative; no forced handling. This is not a core practical.

Book-first evidence: labelled explanation OR observation/result table, original response, true Review edit and each Exit. No invented result table. Label spoken/practical/photo evidence SCI-LAUNCH-A2-W04 on [SYS]; no pupil phones/data in HTML. Code only actual VF/WS/I/NS(+)/E/R///?. Staff prompts separate from I · S · C. Record hinge split, misconception, mode, deepest prompt, independence and actual personal-target link.

40-minute source timing: Arrival 5 + Starter 3 + I do 5 + We do 7 + We do 2 5 + Independent 8 + Review 4 + Exit 3. Resets are INSIDE We do, Independent and Review. Route alternatives share 8; continuation views share their stage budget, never extra time.

If behind retain accurate-model choice/reason, own evidence and safety; after secure hinge shorten rehearsal. Move Exit reflection to Week 5 Arrival only if needed with reason/coverage record. Next exact weekly line: Complete a genetic cross (digital Punnett square). Hide genome label and reconstruct whole-set meaning first.

Fading: full model → missing gene label → independent without answer. Invitation bank/parallel/diagnoses/held ladder/full access controls and mapping are in Staff Answers. [AWARD] [QUAL] Foundation Biology/Combined Science Paper 1 shared 3.5/3.13, no Biology-only/higher-tier assumption. Formative topic quizzes/practical write-ups; December Topics 1–3 assessment. Link E to actual personal targets on [SYS].

Source: Launch/Autumn_2_2026-27/SCI_LAUNCH_A2_W04/SCI_LAUNCH_A2_W04.html; revision c4ca8ffab98e6186a9ef2fed3921644b177f0d6d. Workbook: _authoring/science_2026-27/_toolchain/scheme_references/LAUNCH_SOW.xlsx; LAUNCH Weekly - Autumn B42 Aut2·W4 / C42 Describe DNA, genes & chromosomes. Date _____ Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale